Estimated Annual Agricultural Pesticide Use
Yakima County, Washington 2014-2018

24 July 2026

Summary

Yakima County ranks **#5 of Washington's 39 counties** by total estimated agricultural pesticide mass applied (EPest-high method, summed 2014-2018), making it one of WA's five highest pesticide-use counties. Yakima County is nationally known as the "Fruit Bowl of the Nation," a major tree-fruit region (apples, pears, cherries) and one of the country's leading hop-growing areas.

Table 2 lists the rank and best available estimates of mass applied (kg) for the top 80 agricultural pesticides in **Yakima County, Washington** during 2018, as reported by USGS Estimated Annual Agricultural Pesticide Use data. The table also includes rank by 3-year average (2016-2018) and 5-year average (2014-2018) estimates of mass applied, and each compound's chemical class (abbreviated; see the key in Table 1 and the full classification, with a separate functional class column, in `data/pesticide_classification.csv`, shared with every chapter in this repository). Rankings and estimates are based on the *EPest-high method* described by Thelin and Stone (2013) and Baker and Stone (2015):

> "EPest-low and EPest-high, are used to estimate a range of pesticide use. Both EPest-low and EPest-high methods incorporate proprietary surveyed rates for Crop Reporting Districts (CRDs), but EPest-low and EPest-high estimates differ in how they treat situations when a CRD was surveyed and pesticide use was not reported for a particular crop present in the CRD. In these situations, EPest-low assumes zero use in the CRD for that pesticide-by-crop combination. EPest-high, however, treats the unreported use for that pesticide-by-crop combination in the CRD as missing data. In this case, pesticide-by-crop use rates from neighboring CRDs or CRDs within the same region are used to estimate the pesticide-by-crop EPest-high rate for the CRD." *-Pesticide National Synthesis Project webpage*

USGS lists several caveats of these data, in particular:

- Reliability decreases with scale (e.g. "detailed interpretation of where and how much use occurs within a county is not appropriate").
- EPest-low estimates are more likely to reflect state-based restrictions on pesticide use than EPest-high estimates.
- A blank cell in the source file (no reported/estimated use for that compound-year-county) is treated as zero in this chapter.
- **2019 is excluded from this ranking.** The 2019 county-level file available at the time of writing covers only 69 compounds nationwide, versus roughly 400 compounds in every other year from 1992-2018 — a partial/preliminary release, not a real drop in the number of pesticides in use. Anchoring the ranking on it would silently omit hundreds of real compounds. This chapter uses 2018 (the most recent complete year) as the 1-year snapshot instead; re-run with 2019 -included windows once a complete 2019 release is available.

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)

Code	Chemical class
CPX	Chlorophenoxy compound
DIP	Dipyridyl compound
DTC	Dithiocarbamates
INO	Inorganic compounds
MIC	Microbial
CB	N-methyl carbamates (AChE inhibitor)
OGM	Organo-metallic compound
OC	Organochlorine compound
OP	Organophosphorous compound

Table 1: Chemical class abbreviation key (used in the Class column of Table 2) *(continued)*

Code	Chemical class
OTH	Other
PYR	Pyrethrins/Pyrethroids
TC	Thiocarbamates
TRZ	Triazines

Table 2: Top 80 EPest-high Estimates, Yakima County, WA, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
1	PETROLEUM OIL	OTH	1955166	2067675	1811361
2	SULFUR	INO	496854	411347	372228
3	KAOLIN CLAY	INO	294317	236301	187814
4	CALCIUM POLYSULFIDE	INO	152813	254212	232762
5	COPPER	INO	90285	73620	56646
6	GLYPHOSATE	OTH	57098	73631	65200
7	METAM	DTC	51833	29856	33620
8	CHLORPYRIFOS	OP	39774	40727	42510
9	PARAQUAT	DIP	29162	24350	19490
10	CARBARYL	CB	24910	19619	23723
11	GLUFOSINATE	OTH	23076	10967	8138
12	POTASSIUM BICARBONATE	INO	19983	24219	17939
13	METAM POTASSIUM	DTC	12478	10398	13214
14	ATRAZINE	TRZ	12198	9912	6699
15	TRIFLUMIZOLE	OTH	9544	10090	9805
16	METOLACHLOR & METOLACHLOR-S	OTH	8736	10915	11410
17	METOLACHLOR-S	OTH	8736	10915	11408
18	2,4-D	CPX	7499	13385	17346
19	COPPER HYDROXIDE	INO	7273	10402	7506
20	TRICLOPYR	OTH	7138	4087	3331
21	MANCOZEB	DTC	7109	7964	11556
22	BACILLUS THURINGIENSIS	MIC	6700	6572	4888
23	PENDIMETHALIN	OTH	5989	11614	13191
24	BURKHOLDERIA SPP	MIC	5846	3050	2377
25	NORFLURAZON	OTH	5324	2165	2048
26	DIMETHENAMID & DIMETHENAMID-P	OTH	5299	2351	2023
27	DIMETHENAMID-P	OTH	5299	2351	2023
28	CHLOROPICRIN	OC	5019	2186	2794
29	DICHLOROPROPENE	OC	4826	13631	25902
30	SULFURIC ACID	INO	4479	10304	11885
31	PROHEXADIONE	OTH	4292	2364	1895
32	SIMAZINE	TRZ	4290	1976	3956
33	BUPROFEZIN	OTH	4051	5717	4332
34	PYRACLOSTROBIN	OTH	3743	3217	2878
35	THIOPHANATE-METHYL	OTH	3691	4409	4465
36	CHLORANTRANILIPROLE	OTH	3577	3009	2992
37	MALATHION	OP	3530	2649	2438
38	ACETAMIPRID	OTH	3437	2898	2764
39	SPINETORAM	OTH	3283	2934	3074
40	METRIBUZIN	TRZ	3273	2719	2515
41	FLUOPYRAM	OTH	2969	2589	1935
42	BOSCALID	OTH	2824	3669	4006
43	TRIFLOXYSTROBIN	OTH	2727	2587	2489
44	DIURON	OTH	2576	2892	4184
45	COPPER SULFATE	INO	2321	16472	14363
46	BACILLUS SUBTILIS	MIC	2302	1348	1008

Table 2: Top 80 EPest-high Estimates, Yakima County, WA, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above) *(continued)*

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
47	FLUXAPYROXAD	OTH	2233	1358	860
48	REYNOUTRIA SACHALINENSIS	MIC	2169	797	624
49	PENTHIOPYRAD	OTH	2167	2782	3276
50	SPINOSYN	OTH	2149	2715	2290
51	ETHEPHON	OTH	1999	2296	3085
52	CYPRODINIL	OTH	1885	2061	1636
53	IMIDACLOPRID	OTH	1832	2619	3132
54	SPIRODICLOFEN	OTH	1764	879	867
55	FLUMIOXAZIN	OTH	1520	1306	803
56	FLUTRIAFOL	OTH	1515	1127	964
57	SPIROTETRAMAT	OTH	1478	1783	1498
58	AVIGLYCINE	OTH	1458	1100	844
59	HEXYTHIAZOX	OTH	1356	815	755
60	ORYZALIN	OTH	1310	2811	5278
61	INDAZIFLAM	OTH	1249	1200	1059
62	PHOSMET	OP	1192	6355	6724
63	METHOXYFENOZIDE	OTH	1117	1371	1370
64	EPTC	TC	1058	713	1067
65	AUREOBASIDIUM PULLULANS	MIC	959	1372	902
66	NOVALURON	OTH	920	1005	943
67	DIAZINON	OP	891	1901	2854
68	OXYTETRACYCLINE	OTH	890	993	1256
69	QUINOXYFEN	OTH	871	1452	1177
70	PYRIDABEN	OTH	812	432	651
71	MYCLOBUTANIL	OTH	803	1242	1247
72	CUPROUS OXIDE	INO	800	871	2654
73	DICAMBA	OTH	783	2425	1665
74	PICLORAM	OTH	775	354	317
75	BROMOXYNIL	OTH	730	1155	2804
76	GIBBERELLIC ACID	OTH	720	424	331
77	CHLOROTHALONIL	OTH	677	478	1004
78	DIFENOCONAZOLE	OTH	672	728	583
79	2,4-DB	CPX	662	586	467
80	NAPHTHYLACETIC ACID	OTH	639	527	502

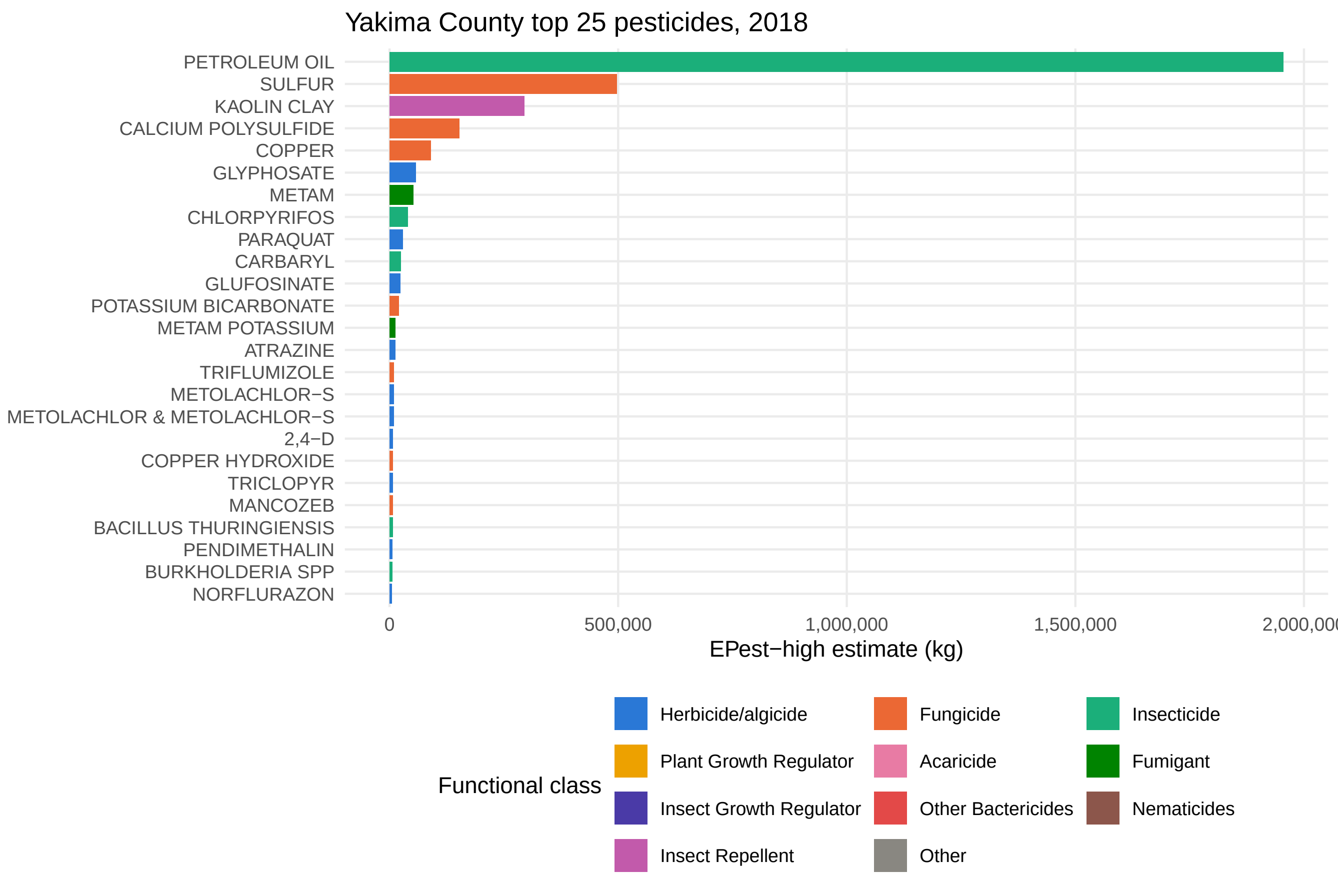

Figure 1: Top 25 compounds in Yakima County by EPest-high mass applied, most recent year, colored by pesticide functional class.

Appendices

Table 3: Top 50 Yakima County Estimates, Range (EPest-low and EPest-high), 1-Yr, 2018

RANK_1YR	COMPOUND	EPEST_LOW_KG_1YR	EPEST_HIGH_KG_1YR
1	PETROLEUM OIL	1955166	1955166
2	SULFUR	496852	496854
3	KAOLIN CLAY	294317	294317
4	CALCIUM POLYSULFIDE	152813	152813
5	COPPER	90285	90285
6	GLYPHOSATE	47093	57098
7	METAM	382	51833
8	CHLORPYRIFOS	39774	39774
9	PARAQUAT	25891	29162
10	CARBARYL	24910	24910
11	GLUFOSINATE	22846	23076
12	POTASSIUM BICARBONATE	19983	19983
13	METAM POTASSIUM	15	12478
14	ATRAZINE	176	12198
15	TRIFLUMIZOLE	9544	9544
16	METOLACHLOR & METOLACHLOR-S	31	8736
17	METOLACHLOR-S	31	8736
18	2,4-D	7340	7499
19	COPPER HYDROXIDE	509	7273
20	TRICLOPYR	0	7138
21	MANCOZEB	6283	7109
22	BACILLUS THURINGIENSIS	6700	6700
23	PENDIMETHALIN	2208	5989
24	BURKHOLDERIA SPP	5819	5846
25	NORFLURAZON	0	5324
26	DIMETHENAMID & DIMETHENAMID-P	5182	5299
27	DIMETHENAMID-P	5182	5299
28	CHLOROPICRIN	0	5019
29	DICHLOROPROPENE	4708	4826
30	SULFURIC ACID	0	4479
31	PROHEXADIONE	4292	4292
32	SIMAZINE	0	4290
33	BUPROFEZIN	4051	4051
34	PYRACLOSTROBIN	3628	3743
35	THIOPHANATE-METHYL	3607	3691
36	CHLORANTRANILIPROLE	3496	3577
37	MALATHION	3530	3530
38	ACETAMIPRID	3437	3437
39	SPINETORAM	3283	3283
40	METRIBUZIN	168	3273
41	FLUOPYRAM	2965	2969
42	BOSCALID	2814	2824
43	TRIFLOXYSTROBIN	2727	2727
44	DIURON	373	2576
45	COPPER SULFATE	2312	2321
46	BACILLUS SUBTILIS	2301	2302
47	FLUXAPYROXAD	2224	2233
48	REYNOUTRIA SACHALINENSIS	0	2169
49	PENTHIOPYRAD	2146	2167
50	SPINOSYN	2147	2149

Table 4: Top 50 Yakima County Estimates, Range (EPest-low and EPest-high), 3-Yr Avg, 2016-2018

RANK_3YR	COMPOUND	EPEST_LOW_KG_3YR_AVG	EPEST_HIGH_KG_3YR_AVG
1	PETROLEUM OIL	2013215	2067675
2	SULFUR	411227	411347
3	CALCIUM POLYSULFIDE	254212	254212
4	KAOLIN CLAY	235844	236301
5	GLYPHOSATE	65899	73631
6	COPPER	73620	73620
7	CHLORPYRIFOS	40372	40727
8	METAM	12676	29856
9	PARAQUAT	22584	24350
10	POTASSIUM BICARBONATE	24219	24219
11	BACILLUS AMYLOLIQUEFACIEN	18231	20057
12	CARBARYL	19415	19619
13	COPPER SULFATE	16446	16472
14	DICHLOROPROPENE	13552	13631
15	2,4-D	13115	13385
16	ZINC	12878	12878
17	PENDIMETHALIN	8387	11614
18	GLUFOSINATE	9424	10967
19	METOLACHLOR & METOLACHLOR-S	71	10915
20	METOLACHLOR-S	71	10915
21	COPPER HYDROXIDE	8094	10402
22	METAM POTASSIUM	5	10398
23	SULFURIC ACID	0	10304
24	TRIFLUMIZOLE	10090	10090
25	ATRAZINE	84	9912
26	MANCOZEB	7671	7964
27	ZIRAM	6597	6597
28	BACILLUS THURINGIENSIS	6572	6572
29	PHOSMET	6355	6355
30	BUPROFEZIN	5717	5717
31	THIOPHANATE-METHYL	4312	4409
32	TRICLOPYR	0	4087
33	BOSCALID	3311	3669
34	PYRACLOSTROBIN	2932	3217
35	BURKHOLDERIA SPP	2829	3050
36	CHLORANTRANILIPROLE	2976	3009
37	SPINETORAM	2934	2934
38	ACETAMIPRID	2898	2898
39	DIURON	426	2892
40	ORYZALIN	2811	2811
41	PENTHIOPYRAD	2754	2782
42	METRIBUZIN	1255	2719
43	SPINOSYN	2711	2715
44	MALATHION	2644	2649
45	IMIDACLOPRID	2615	2619
46	FLUOPYRAM	2430	2589
47	TRIFLOXYSTROBIN	2501	2587
48	CHROMOBACTERIUM SUBTSUGAE	0	2451
49	BIFENTHRIN	40	2446
50	DICAMBA	442	2425

Table 5: Top 50 Yakima County Estimates, Range (EPest-low and EPest-high), 5-Yr Avg, 2014-2018

RANK_5YR	COMPOUND	EPEST_LOW_KG_5YR_AVG	EPEST_HIGH_KG_5YR_AVG
1	PETROLEUM OIL	1778685	1811361
2	SULFUR	370788	372228
3	CALCIUM POLYSULFIDE	232631	232762
4	KAOLIN CLAY	186923	187814
5	CAPTAN	125	98350
6	GLYPHOSATE	59897	65200
7	COPPER	56642	56646
8	CHLORPYRIFOS	42014	42510
9	METAM	15216	33620
10	DICHLOROPROPENE	25500	25902
11	CARBARYL	23495	23723
12	PARAQUAT	17809	19490
13	PINOLENE	0	18606
14	POTASSIUM BICARBONATE	17824	17939
15	2,4-D	13327	17346
16	BACILLUS AMYLOLIQUEFACIEN	10938	14596
17	COPPER SULFATE	13664	14363
18	METAM POTASSIUM	5126	13214
19	PENDIMETHALIN	10759	13191
20	DODINE	0	12999
21	SULFURIC ACID	0	11885
22	MANCOZEB	10796	11556
23	METOLACHLOR & METOLACHLOR-S	99	11410
24	METOLACHLOR-S	99	11408
25	TRIFLUMIZOLE	9805	9805
26	ZIRAM	9247	9247
27	ACEPHATE	0	8584
28	GLUFOSINATE	5689	8138
29	COPPER HYDROXIDE	6102	7506
30	PHOSMET	6724	6724
31	ATRAZINE	72	6699
32	ORYZALIN	4654	5278
33	ZINC	5237	5237
34	BACILLUS THURINGIENSIS	4888	4888
35	THIOPHANATE-METHYL	3211	4465
36	BUPROFEZIN	4332	4332
37	DIURON	1621	4184
38	BOSCALID	3777	4006
39	SIMAZINE	2924	3956
40	TRICLOPYR	1	3331
41	PENTHIOPYRAD	3253	3276
42	IMIDACLOPRID	3092	3132
43	ETHEPHON	3085	3085
44	SPINETORAM	3066	3074
45	HYDROGEN PEROXIDE	659	3018
46	CHLORANTRANILIPROLE	2961	2992
47	PYRACLOSTROBIN	2633	2878
48	DIAZINON	2829	2854
49	BROMOXYNIL	788	2804
50	CHLOROPICRIN	1500	2794